Research and Realization on Improved MANET Distance Broadcast Algorithm Based on Percolation Theory

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***Abstract*—Percolation theory coincided with MANET model can be used for MANET network connectivity problems, and it is a new mobile ad hoc network topology control strategy. We propose the application of percolation theory to improve distance broadcast algorithm of mobile ad hoc network, the node to the antenna, only in specific sectors related to the direction of forwarding, realize the directed forwarding of the broadcast message; We use additional coverage area instead of the distance threshold of traditional distance broadcast algorithm as the critical probability threshold that the network is just connected, which improves the network broadcast forwarding rate, reduces the end-to-end delay, achieves the node energy effective utilization, and therefore attains the purpose of the network energy-saving. Through the simulation experiment, the packet delivery rate and average delay are improved greatly compared with the traditional distance broadcast algorithm.**

Keywords-Percolation Theory; MANET; Distance Broadcast

I. CHARACTERISTICS OF MANET

MANET is composed of a group of both terminals and routing function mobile devices via a wireless link forming a multi-hop temporary autonomous system. It does not require any central control, network can be automatically detected and integrated the new node; When any node is away from the network, the remaining nodes can also automatically reconfigured to adapt to the new scenes. In the absence or the existing network infrastructure failure cases, it can provide communication among terminals. What's more, MANET has important applications in the military, agriculture and emergency rescue.

II. MANET TOPOLOGY CONTROL STRATEGIES AND METHODS

MANET nodes can be moved at any speed and way, but power of wireless transmitting device randomly changes and the wireless channel interfere with each other and so on, the communication state after the formation of the topology is always variable and unpredictable. If there is no topology control, all nodes will work with the greatest power and the energy of the node will be rapid consumed by communication part reducing the network life cycle, resulting in wireless signal frequently conflicts, influencing of node communication quality, reducing the throughput of the network, which can also cause routing computation more complex, consume more resources. Therefore, the MANET for effective topology control [1, 2] is very necessary.

A. MANET Topology Control Strategy

To make a network topologies satisfy the following properties: Connectivity. In order to realize the nodes can communicate with each other, we must guarantee the connectivity of the generated topology. That is we can send a message to another node from any one node [3].

1) Symmetry. Hypotheses from node i to node j there is an side, you can also find another side from node j to node I. That is the generated topology in the series circuit is symmetrical.

2) Sparse. Generate the topology of edge number O (n), where n is the number of nodes. Reducing the number of edges of topology can effectively reduce the network interference, improve the network throughput, but also simplify the routing host, save valuable resources.

3) Plane. In the generated topology there are no edges intersection. According to the graph shows, meeting the plane must meet the sparsity.

4) Node bounded degree. The generation of topological node in the neighbor number is less than a constant d. Reduce node degrees can reduce the number of node forwarding the message while reducing routing computation complexity.

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5) Spanner properties. In the generated topology, the distance of any two nodes is less than the constant times in the undirected graph.

B. MANET Topology Research Methods

Topological research methods are mainly divided into two classes, Geometric method and probability method. Geometry method is, according to certain geometric structure based on the topology of the network, to satisfy certain properties. Probability method is, under the nodes in accordance with certain probability density distribution ,to calculate the minimum transmission power and the minimum number of neighbors the node desired when topology probability satisfies certain properties. However, the development of mature random graph theory is not suitable for MANET. Random drawing assumptions does not accord with the features of MANET between two nodes independent of each other.

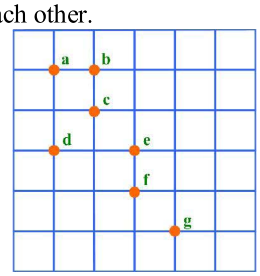

Figure 1. Point Percolation Model

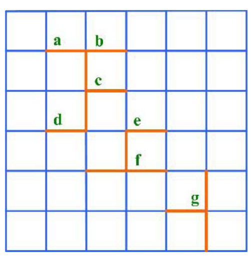

Figure 2. Bond Percolation Model

Based on the probability of MANET broadcast algorithm[4], percolation theory[5] of two models of forwarding probability for network connectivity critical threshold probability Pc, in other broadcast algorithm, network connectivity critical probability threshold is equivalent to its own properties and related parameters. Application of percolation theory model based on distance MANET broadcast algorithm is to find the forward probability, network connectivity critical probability threshold is equivalent to and their related to the nature of the distance parameter threshold.

III. PERCOLATION THEORY AND MANET NETWORK TOPOLOGY

A. Percolation theory and MANET network topology mapping

Simply, the percolation is the flow of fluid through porous media. Porous media [6] refers to the solid material containing voids, cracks and other types of capillary system medium. From one side to the other side of the medium continuous channel, and the pores and channels throughout the medium has a wide distribution.

There are three characteristics of porous medium. First, a porous medium (or porous material) is a multiphase medium to occupy a space, including solid phase as part of the solid skeleton, without being solid section occupied space known as the pore. Pores can be a gas or liquid, can also be a multiphase fluid; Second, solid and porous should be throughout the medium, that is in the medium in a proper size of the body element, the element must have a certain proportion of the solid particle and pore; Third, there should be some or most other connected, and the fluid can flow in it, this part of the pore space is called the effective pore space, unconnected pore space or connected the dead end pores in this part of the space is invalid for the pore space, fluid through porous flow, actually can be considered as solid skeleton.

MANET node in a two-dimensional spatial random distribution, MANET network constitutes a two-dimensional space can be mapped to a porous medium, the MANET node mapping for porous media of the solid skeleton, and the two-dimensional space of the other part is corresponding to the pore, the packet forwarding is mapped to the flow of fluid in porous.

The core of MANET distance broadcast algorithm based on percolation theory is that nodes reach critical probability threshold and network instantaneous get connectivity, forming a node connected with each other and the infinite order component, then mapping for the effective pore space. Therefore the percolation theory model is consistent with the MANET model, application of percolation theory can be used to analyze the MANET connectivity, and the improved MANET algorithm based on probability broadcast also has a prominent role.

B. Percolation Theory State Mutations Mapped MANET Network Connectivity Critical Threshold.

From a mathematical point of view, percolation theory often focus on the network topology is the grid model, percolation theory mainly studies critical state mutations (Phase Transition) [7].

In the one-dimensional percolation model, this phenomenon of mutation status does not occur significantly.

In the two-dimensional percolation model, percolation model is divided into two categories, one is the point of percolation (Site Percolation) model, and the other is side percolation .

First, the definition given in two-dimensional percolation model, define a grid, which consists of N points and M edges, this lattice rules generally do not require it to be standard, but the actual research is often used the type rules.

Fig.1 point percolation model. Let one point G is occupied with a certain probability P, and is not occupied with the probability of 1-P. If the enough occupied points will be adjacent to each other, they form a group or as clusters. Any occupied point in cluster must be contiguous to at least one point, and occupied by a finite number of points to form a group called finite groups (clusters). Occupied by the infinite number of points to form a group called point percolation group. An isolated point can be accounted for as a group with only one point. Two points away from infinity can be connected by edges, this group is called bond percolation group.

Fig.2 bond percolation model. Let one edge G is occupied with a certain probability P, and is not occupied with the probability of 1-P.Now occupied and painted on a thick edge which connected to each other, we said these edges formed a bond group or called bond clusters. Any occupied edge in cluster must be adjacent to at least one edge, and occupied by a finite number of edges to form a group called finite groups (clusters). If a group contains infinite edges, in two-dimensional percolation model, the mutation phenomenon occurs in the value of probability P. The probability of state mutation P-value denoted Pc, and also known as the critical point or critical threshold. Less than the probability of Pc, the system is called sub-critical state and global state mutation does not occur; Greater than the probability of Pc, the system is called super-critical state and global state must change just.

IV. IMPROVED DISTANCE MANET BROADCAST ALGORITHM BASED ON THE PERCOLATION THEORY

As an additional coverage area EAC and the distance D of two nodes are a linear relationship, in order to calculate more accurately, EAC can be used instead of the distance D as the critical probability threshold.

So, when the EAC is greater than the critical threshold, the node forwards the broadcast message. The application of percolation theory based on the improved distance broadcast algorithm, node is equipped with a directional antenna, AOA (Angle of Arrival cross location) technology can be applied to acquire the direction of the node to receive broadcast messages. If the distance D is known, utilizing the angle of arrival (AOA Angle of Arrival crosses location) technology and distance D compute node value of each sector in the EAC.

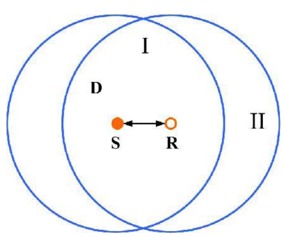

Figure 3. Directive EAC

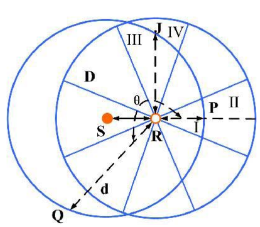

Figure 4. Omnidirectional EAC

Coverage of the receiving node is divided into eight sectors, then split the eight sectors, identifies the intersection of coverage area of 8 bisectors and the source node. Set variable d, all of these lines being made up of the receiving node and the 8 nodes can be the variable d.

Fig.3, II part shows the additional coverage of omni-directional antenna between the source node and the forwarding node.

Fig.4, S is the source node, R is the forwarding node, node R uses a directional antenna, the transmission range is divided into eight sectors, each sector and then divided equally, intersection of bisectors among the three sectors and source node S coverage are T, P, Q. If d is a variable, RT, RP, RQ value can be the value of d, R is the transmission radius of node, as shown in the figure angle θ is known, assuming RQ = d, the explanation of solution process of the d value is showed in fig.5:

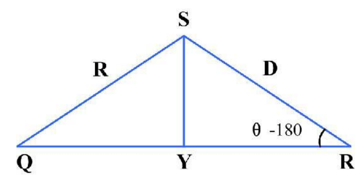

Figure 5. solution of d

A. With S vertices at QR vertical line, crossing of the QR with Y. According to the law of cosines.

$$YR = COS\ (\theta - 180) \bullet D = -D \bullet COS\ \theta \tag{1}$$

B. According to the sine law

$$SY = \sin(\theta - 180) \bullet D = -D \bullet \sin\theta \tag{2}$$

C. According to the Pythagorean theorem

$$QY = \sqrt{R^2 - (-D \bullet \sin\theta)^2} = \sqrt{R^2 - (D \bullet \sin\theta)^2} \tag{3}$$

D. Get

$$d = QY + YR = \sqrt{R^2 - (D \bullet \sin\theta)^2} - D \bullet \sin\theta \tag{4}$$

RT, RP are corresponding to the additional coverage area of II and IV. When the RP <RT, Ⅳ <Ⅱ. The smaller the additional coverage, the lower the necessity for forwarding, forwarding probability Pi is also smaller. Defining the forward probability Pi is calculated by the following formula:

$$P_i = \max\{0, 1 - (\frac{d}{R})^2\} \tag{5}$$

Application of d and D recalculate the value of Pi, additional coverage will evolve as the distance d, and the forwarding probability Pi associated with distance d, forwarding nodes in different sectors have different forwarding probability Pi, thus which can better adapt to network density. Regional and sector coverage of the source node transmitting are the more overlap, d value is larger, on the contrary forwarding probability Pi is the smaller, or even zero, then give up the sector direction forwarding. Take QR = d for example, the d > R, then d / R > 1, then according to the formula(5), Pi is 0, that is, give up forwarding broadcast messages. This is also satisfied with the assume that the EAC is too small to forwarding.

V. Algorithm Simulation

Design follows the simulation scheme: in a 250 x 250 regional randomly distributed within100 nodes, the link layer based on 802.11 standard MAC protocol, the MAC protocol using CSMA / CD multiple access mode, the maximum node transmission range of 250m, the maximum speed for 20m / s. Consider the 1 jump CBR business flow, CBR business flow transmission rates per second were 2 packets, 4 packets, 6 packets, 8 packets, 10 packets. The simulation time is 100 seconds, the antenna node transmission range is divided into 8 sectors, and distance threshold values of the omnidirectional node antenna were 60,120. It proved that percolation theory improved distance broadcast algorithm is greatly better with the result of data packet delivery rate and the average end-to-end delay of data packet parameters than the traditional distance broadcast algorithm.

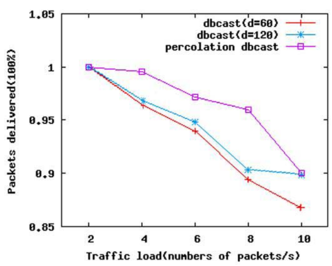

Figure 6. Packets Delivered Contrast

As shown in Fig.6, we establish data packet delivery rate comparison chart for the different service loads, dbcast (d = 60) are the data packet delivery rates as distance threshold value of 60 in traditional distance broadcast algorithm, dbcast (d = 120) are as distance threshold value of 120 in traditional distance broadcast algorithm. Percolation dbcast are improved MANET distance broadcast algorithm based on percolation theory. From figure 6, along with the service load increasing, data packet delivery rate of improved MANET distance broadcast algorithm is better than the traditional distance broadcast algorithm.

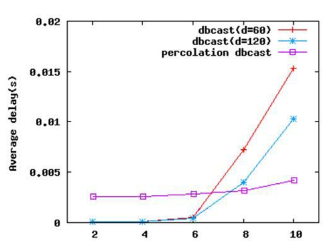

Figure 7. Average Delay Contrast

As shown in Fig.7, it is load average delay comparison chart for the different traffics, dbcast (d = 60) are average delay as the distance threshold value of 60, dbcast (d = 120) are as the distance threshold value of 120. Percolation dbcast are average delay of improved MANET distance broadcast algorithm based on percolation theory. As you can see from figure7, when the load is relatively lower, average delay time is very smaller, with the load increasing, average delay of improved MANET distance broadcast algorithm are significantly less than the conventional distance broadcast algorithm.

VI. Conclusion

Improved MANET distance broadcast algorithm based on percolation theory are to select a particular direction forwarding with extra area EAC instead of distance threshold as network connectivity mutation forwarding a critical threshold, and network can be set up more accurate transmission power, radically reduce the network interference, improve the throughput of the network, saving resource nodes, prolong the network life cycle.

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